

feedback circuitry to clamp the output voltage to a constant value by either bucking (lowering) or boosting (increasing) the input voltage, depending on its level.

The principal element of a stabiliser is a transformer whose output voltage is controlled depending on the input voltage. It controls the output voltage between the low and high voltage thresholds at which the device operates, but if these limits are surpassed, it simply switches off. A stabiliser cannot regulate the frequency of the output voltage or provide any form of backup power, but does provide adequate control of voltage for most applications in case of occasional low-level fluctuations.

■ **UPS:** This is the most effective way to implement power protection. A UPS is a device capable of maintaining pure AC power and can even provide backup power in case of very poor input power or a total power failure. Here, DC power provided by the UPS batteries is converted into AC power by an

electronic circuit called an inverter. This gives clean power to the equipment connected to it.

There are three types of UPS systems in use—offline, online and line-interactive. In an offline UPS, the equipment is driven directly by the mains power as long as it remains within threshold limits of the minimum and maximum values. During this time, the batteries of the UPS are continually charged. In case of an extreme power condition or a power failure, the UPS switches on very quickly (usually under 5 milliseconds) to the batteries.

In an online UPS, the equipment is continually driven through the batteries even during normal conditions, during which time the batteries also charge. Therefore, this type of UPS understandably provides the purest form of power protection available—no matter what the input power conditions, the output power is provided through the battery.

## The ultimate protection

**A**nti-virus protection is dandy, but how do you safeguard against losing valuable data to a black out? Prevention is better than cure, invest in a suitable power protection system to safeguard your equipment and data.

Before you buy a UPS, you need to add the total power consumption of your system. Decide which pieces of equipment need UPS support. Typically, only the CPU and monitor are supported, this reduces the power drawn from the UPS.

If the power draw is expressed in Amps, then multiply that value by your nominal line voltage (230V). If the power draw is expressed in Watts, multiply by a factor of 1.3 to 1.4 for VA load. For example, 250 Watt power supply (x 1.4) = 350 VA load (assuming that your monitor is drawing power from the power supply of the PC). For redundancy and safety, a 500 VA UPS should be enough for most PCs.

### Technology Insight

Here's how the UPS system works and keeps your computer up and running. The rectifier or the charger converts the incoming Alternating Current (AC) mains into Direct Current (DC), which is used to charge the battery, and in some cases to power the inverter. Then the inverter takes DC power from the rectifier or the battery and converts it into AC power for the load. Most UPS systems include additional filters to eliminate spikes.

### Smart Tips

**Input voltage range:** This is the range beyond which the UPS delivers power through its batteries. The greater the range the UPS can handle, the more versatile its power protection. Look for a voltage range between 150 and 270 volts.

**VA rating:** This is a rating of the

power that can be provided by the UPS. You should choose the rating on the basis of the power requirements of the equipment you are going to connect to the UPS. As a rule of thumb, a 1 GHz processor-based machine with a 20 GB hard disk, a CD-ROM drive and a 17-inch colour monitor would require a 500 VA UPS.

**Backup time:** This is the specified time for which the UPS provides power from the batteries for a fixed load. Look for a backup time of at least 10 minutes at maximum load.

**Switching/transfer time:** Used interchangeably, switching or transfer time is the time taken by the UPS to switch from normal delivery power mode to battery mode when it detects power anomalies. Look for transfer times of less than 5 milliseconds.

**Alarms and connectivity:** Most UPS systems have LEDs on the front panel that indicate the state of its batteries and the condition of the input and output power. This visual indication of its operation is important especially during power failures—look for a UPS that features these indicators.

**Charging time:** This is the time taken to charge the UPS batteries from a completely drained state to a fully charged condition. This should be between 4 and 6 hours.



### Decision Maker

|          | Single computer protection                                                                                                                                                                                                       | Mission-critical applications                                                                                                                                                                                                            |
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| You Need | Protection for a single computer comprising a monitor, system unit, keyboard and mouse, for example, a 1 or 2 GHz processor-based machine with a 20 GB hard disk, a CD/DVD-ROM drive, 256 to 512 MB of RAM and a 17-inch monitor | Protection for a server that needs to be continually running with mission-critical applications, for example, a 1 GHz-processor, with up to 1 GB of RAM, up to four SCSI hard disks, a CD-ROM drive, backup device and a 17-inch monitor |
| Look For | An offline or line-interactive UPS with a 450 VA to 650 VA rating, and a backup time of 10 minutes                                                                                                                               | An online UPS with at least 1 KVA rating, and a backup time of 10 minutes. It should also support remote monitoring through a serial port with alarms and indicators in case of power anomalies                                          |